

# Security+ 3.4 · Recovery Testing

Mestre CyberSec · Lesson 3.4 — Recovery Testing (5:18) · [Assistir Vídeo no YouTube](#) ↗ · Treino Interativo: [edusouzamarques.github.io/cybersec](https://edusouzamarques.github.io/cybersec)

**Executive Lesson Summary:** Teste o plano de recovery ANTES do desastre: restaura backup, simulada falha. RPO (quanto dato perde) e RTO (quanto tempo pra voltar). Table-top / parallel / full failover.

## 💡 Key Exam Highlights & Critical Concepts

- RPO: quantos dados podes perder; RTO: quando volta
- Plano NÃO-testado = plano que NÃO funcionará

## 🎯 Exam Tips & Typical CompTIA Scenarios

- 'maximum data loss in hours'=RPO
- 'maximum downtime'=RTO

## Beyond the Video: Official Exam Concepts, Protocols & Traps (Section 3.4 — Recovery Testing)

**Why is this advanced coverage required for SY0-701?** Video lectures provide foundational concepts, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

### • Recovery Point Objective (RPO)

The maximum acceptable amount of data loss measured in time, indicating how far back a recovery must restore data after an incident.

⚠ **Exam Trap:** Gatilho clássico: 'maximum data loss in hours/tempo' = RPO. Não confunda com RTO, que se refere ao tempo de parada/downtime.

### • Recovery Time Objective (RTO)

The maximum acceptable downtime or time allowed to restore a system, service, or process after a disruption.

⚠ **Exam Trap:** Gatilho clássico: 'maximum downtime' = RTO. Não confunda com RPO, que mede perda de dados, não tempo de indisponibilidade.

### • Tabletop Exercise (TTX)

A discussion-based recovery test where stakeholders walk through a disaster scenario to validate roles, communications, and decision-making without changing production systems.

⚠ **Exam Trap:** Baixo custo e baixo risco; valida processos e comunicação, mas NÃO testa tecnicamente a restauração de backups ou o failover real.

### • Parallel Processing (N/A)

A recovery test where the alternate/recovery site runs in parallel with the primary site for a period, allowing processing to continue while validating recovery capabilities.

⚠ **Exam Trap:** Não interrompe a produção e testa o ambiente alternativo, mas não é um failover completo; se a questão exigir transferência total, escolha full failover.

### • Full Failover (N/A)

A recovery test in which the primary site is shut down and all operations are transferred to the alternate/recovery site, providing the most realistic test of recovery readiness.

⚠ **Exam Trap:** Mais realista e mais disruptivo; use quando a pergunta indicar 'actual failover', 'full transfer' ou 'primary site down'.

### • Backup Restoration (N/A)

The process of recovering data from a backup to verify that the backup is readable, complete, and capable of restoring operations within RPO/RTO requirements.

⚠ **Exam Trap:** Backup não testado não é confiável; a restauração comprova a recuperabilidade dos dados e ajuda a validar se o RPO pode ser cumprido.

### • Recovery Plan Testing (N/A)

Scheduled validation of a business continuity/disaster recovery plan using reviews, tabletop exercises, simulations, parallel tests, or full failovers to identify gaps before a real disaster.

⚠ **Exam Trap:** Plano não testado = plano que provavelmente não funcionará. Deve ser repetido após mudanças significativas na infraestrutura ou nos processos.

### • Business Impact Analysis (BIA)

A process that identifies critical business functions and determines the impact of their disruption, helping set recovery objectives such as RTO, RPO, and MTD.

⚠ **Exam Trap:** A BIA antecede e orienta o plano de recuperação; não é um teste técnico, mas define criticidade e objetivos como RTO/RPO.

## ✓ Authentic Practice Questions (10 Real Exam Scenarios)

Scenario-based exam questions. Complete explanations and distractor analysis are provided in the answer key below.

### QUESTION 1 · EXAM SCENARIO

A payment processing company reviews its disaster recovery requirements. Executives state that the company can tolerate losing no more than 10 minutes of transaction data after a disruption, but it may remain offline for up to four hours. Which recovery objective is represented by the 10-minute limit?

- A) Recovery time objective (RTO)
- B) Maximum tolerable downtime (MTD)
- C) Recovery point objective (RPO)
- D) Mean time to repair (MTTR)

### QUESTION 2 · EXAM SCENARIO

A hospital's disaster recovery team is defining how quickly its electronic health record system must be available after an outage. Clinical leadership says the system must be restored within two hours to avoid patient safety issues. Which metric should the team document?

- A) Recovery point objective (RPO)
- B) Recovery time objective (RTO)
- C) Service level agreement (SLA) for backups
- D) Business impact analysis (BIA) threshold

### QUESTION 3 · EXAM SCENARIO

A financial services firm wants to validate its ransomware recovery plan without affecting production systems. Managers, legal, communications, and IT staff meet in a conference room to discuss roles, decisions, and communication steps during a simulated ransomware event. Which type of recovery test is being performed?

- A) Full failover
- B) Parallel processing
- C) Backup restoration test
- D) Tabletop exercise (TTX)

### QUESTION 4 · EXAM SCENARIO

An e-commerce company wants to verify that its secondary data center can handle the full production workload. The primary site remains online and serves customers, while the secondary site processes a copy of the workload for a few hours. Which recovery testing approach is described?

- A) Full failover
- B) Tabletop exercise
- C) Parallel processing
- D) Simulation using backup tapes

#### QUESTION 5 · EXAM SCENARIO

During a scheduled disaster recovery test, a retailer redirects all customer traffic from its primary data center to its secondary data center and shuts down the primary site for two hours. Which type of test did the retailer perform?

- A) Parallel processing
- B) Tabletop exercise
- C) Backup restoration
- D) Full failover

#### QUESTION 6 · EXAM SCENARIO

A database administrator restores the latest backup of a payroll system to an isolated network, verifies record counts and file integrity, and confirms the application can start. No production users are redirected. Which recovery testing activity is being performed?

- A) Full failover test
- B) Backup restoration test
- C) Tabletop exercise
- D) Business impact analysis

#### QUESTION 7 · EXAM SCENARIO

Before setting recovery priorities, a manufacturing company identifies its most critical business processes, the systems they depend on, and the impact of downtime and data loss over time. Which activity is the company performing?

- A) Vulnerability assessment
- B) Full failover test
- C) Business impact analysis (BIA)
- D) Backup restoration test

#### QUESTION 8 · EXAM SCENARIO

A company's recovery plan states that its customer database has an RPO of 15 minutes. However, the database is backed up once every 24 hours. A failure occurs late in the business day. Which recovery objective is most likely not being met?

- A) Recovery time objective (RTO)
- B) Mean time between failures (MTBF)
- C) Maximum tolerable downtime (MTD)
- D) Recovery point objective (RPO)

#### QUESTION 9 · EXAM SCENARIO

A disaster recovery test shows that the order management system can be recovered in six hours. Business leaders previously documented that the system must be restored within four hours. Which metric was not satisfied by the test?

- A)** Recovery point objective (RPO)
- B)** Recovery time objective (RTO)
- C)** Service level objective (SLO) for uptime
- D)** Business impact analysis (BIA) likelihood rating

#### QUESTION 10 · EXAM SCENARIO

A cloud operations team successfully restores encrypted backups to a recovery environment, but the restored application cannot read the data because the required keys are unavailable. Which aspect of recovery testing should the team improve?

- A)** Increasing the frequency of full failover tests
- B)** Reducing the RTO by purchasing faster storage only
- C)** Verifying that encryption keys and dependent secrets are available during restoration
- D)** Replacing the BIA with a vulnerability scan

## ANSWER KEY & COMPREHENSIVE DISTRACTOR EXPLANATIONS

### **Question 1: Option C — Recovery point objective (RPO)**

The RPO defines the maximum acceptable amount of data loss, measured in time before the incident. A 10-minute data-loss limit is therefore an RPO. RTO is the maximum acceptable downtime, MTD is the total time a business can tolerate an outage, and MTTR is the average time to repair a failed component.

### **Question 2: Option B — Recovery time objective (RTO)**

RTO specifies the maximum acceptable time to restore a system or process after a disruption. A two-hour restoration requirement is an RTO. RPO concerns data loss, an SLA is a broader contract, and a BIA threshold may inform RTO but is not the metric itself.

### **Question 3: Option D — Tabletop exercise (TTX)**

A tabletop exercise is a discussion-based test where stakeholders walk through a scenario to validate plans, roles, and decision-making without touching live systems. Full failover and parallel processing involve actual systems, while backup restoration tests focus on recovering data.

### **Question 4: Option C — Parallel processing**

Parallel processing runs the recovery environment alongside the primary environment. The secondary system processes the workload while the primary remains active, allowing validation without cutting over users. Full failover would move production to the secondary site.

### **Question 5: Option D — Full failover**

A full failover test moves actual production operations to the recovery site and may temporarily shut down the primary site. This provides the most realistic validation but carries higher risk. Parallel processing would keep the primary active, tabletop is discussion-based, and backup restoration only validates data recovery.

### **Question 6: Option B — Backup restoration test**

A backup restoration test validates that backups can be successfully restored and used. Performing it in an isolated network reduces risk. Full failover redirects production, tabletop is discussion-based, and BIA identifies business dependencies rather than restoring data.

### **Question 7: Option C — Business impact analysis (BIA)**

A BIA identifies critical business functions, dependencies, and the impact of disruption. Its results help determine RTO and RPO values. Vulnerability assessment finds weaknesses, full failover tests recovery capabilities, and backup restoration validates data recovery.

### **Question 8: Option D — Recovery point objective (RPO)**

An RPO of 15 minutes means no more than 15 minutes of data should be lost. A once-daily backup can result in up to nearly 24 hours of data loss, so the RPO is not met. RTO concerns downtime, MTBF concerns reliability, and MTD concerns total outage tolerance.

### **Question 9: Option B — Recovery time objective (RTO)**

The RTO is the maximum acceptable time to restore operations. The test recovered the system in six hours, exceeding the four-hour RTO. RPO concerns acceptable data loss, an SLO may be related but is not the recovery-specific metric here, and BIA likelihood is not a recovery time target.

### **Question 10: Option C — Verifying that encryption keys and dependent secrets are available during restoration**

Backup restoration testing must include not only the backup data but also the keys, credentials, certificates, and secrets needed to use it. If keys are unavailable, recovery fails even when backups restore. The other options do not address the missing key dependency.